AI Communication Systems Study Kit

Glossary and Practice Quiz — AM/FM Simulation

PART A — GLOSSARY

These definitions cover the main terms used in the project's AM/FM simulation. They describe standard communication-system concepts.

Term	Meaning
Communication system	A system that transfers information from a source to a destination through a channel.
Message signal	The signal containing the information that needs to be transmitted.
Carrier signal	A usually higher-frequency signal whose parameter is varied to carry the message.
Modulation	The process of varying a carrier signal's parameter according to the message signal.
Amplitude	The magnitude or height of a signal measured relative to its reference level.
Frequency	The number of cycles of a periodic signal per second, measured in hertz (Hz).
Time period	The time taken for one complete cycle; for a periodic signal, T = 1/f.
Amplitude Modulation (AM)	A modulation method in which the carrier amplitude varies according to the message signal.
Frequency Modulation (FM)	A modulation method in which the carrier's instantaneous frequency varies according to the message signal.
Modulation index (AM)	For conventional single-tone AM, μ indicates the modulation depth; μ = message amplitude / carrier amplitude under the standard model.
FM modulation index (β)	For a single-tone FM signal, β is the peak frequency deviation divided by the message frequency.
Sampling frequency	The number of samples taken per second when representing a continuous-time signal digitally.
Waveform	A graphical representation showing how a signal varies over time.
Envelope	The smooth outline of an AM waveform that follows the message signal when conventional AM is not overmodulated.
Noise immunity	A system's ability to maintain useful information in the presence of unwanted disturbances.
NumPy	A Python library commonly used for numerical arrays and mathematical calculations.
Matplotlib	A Python library used to create plots and graphs.
Google Colab	A browser-based environment for writing and running Python notebooks.

PART B — PRACTICE QUIZ

Try answering the questions before checking the answer key at the end.

1. What is the main purpose of modulation?

A. To vary a carrier parameter according to the message

B. To remove all frequencies from a signal

C. To convert every signal into DC

D. To eliminate the need for a channel

2. In conventional AM, which carrier property varies with the message signal?

A. Sampling frequency

B. Amplitude

C. Time axis

D. Carrier's nominal frequency

3. In ideal FM, which property varies with the message signal?

A. Carrier amplitude only

B. Sampling rate only

C. Instantaneous frequency

D. Plot line thickness

4. What does Hz measure?

A. Signal height

B. Cycles per second

C. Time per sample only

D. Modulation depth

5. Which Python library is commonly used in this code to draw graphs?

A. Matplotlib

B. pathlib

C. getpass

D. zipfile

6. What does NumPy help with in the simulation?

A. Physical antenna alignment

B. Numerical arrays and calculations

C. Internet connectivity

D. PDF page design

7. True or False: In ideal single-tone FM, the carrier amplitude remains constant.

A. True

B. False

8. True or False: In conventional AM, the message is represented by changing the carrier frequency.

A. True

B. False

9. What is a waveform?

A. A graph showing a signal's variation over time

B. A list of programming errors

C. A type of computer keyboard

D. A measure of file size

10. Why are AM and FM graphs useful in a study kit?

A. They replace all theory permanently

B. They make signal behavior easier to visualize and compare

C. They guarantee zero noise in transmission

D. They prove an AI model is present

PART C — ANSWER KEY WITH EXPLANATIONS

1. What is the main purpose of modulation? Answer: A

Modulation changes a carrier parameter in accordance with the information signal.

2. In conventional AM, which carrier property varies with the message signal? Answer: B

AM varies carrier amplitude; the carrier frequency is normally fixed.

3. In ideal FM, which property varies with the message signal? Answer: C

FM changes the carrier's instantaneous frequency while ideal amplitude remains constant.

4. What does Hz measure? Answer: B

One hertz equals one cycle per second.

5. Which Python library is commonly used in this code to draw graphs? Answer: A

Matplotlib provides plotting functions such as plot() and title().

6. What does NumPy help with in the simulation? Answer: B

NumPy supports arrays, time samples, cosine functions, and numerical calculations.

7. True or False: In ideal single-tone FM, the carrier amplitude remains constant. Answer: A

Ideal FM varies instantaneous frequency, not the carrier amplitude.

8. True or False: In conventional AM, the message is represented by changing the carrier frequency. Answer: B

That describes frequency variation; conventional AM changes carrier amplitude.

9. What is a waveform? Answer: A

A waveform displays signal amplitude or another signal quantity against time.

10. Why are AM and FM graphs useful in a study kit? Answer: B

Plots help learners connect equations with visible signal behavior; they do not alone prove AI functionality.

Project accuracy note

This glossary and quiz explain standard AM/FM concepts and the Python tools mentioned in the sample simulation. They do not independently verify the user's project code. Describe an AI feature only if it is actually implemented and demonstrable in the project.